

Recap

The n 'th term of a sequence $\{a_n\}$ is an :

- Arithmetic sequence
 - $a_n = a + dn$
- Geometric Sequence
 - $a_n = ar^n$

Recursive Definitions

Another way to build a sequence is with a rule that lets us compute one new term at a time,

Example:

If $a_0 = 1$ and $a_{n+1} = a_0 + a_1 + a_2 + \dots + a_n$ when $n \geq 0$

We get:

$$a_1 = a_0 = 1$$

$$a_2 = a_0 + a_1 = 1 + 1 = 2$$

$$a_3 = a_0 + a_1 + a_2 = 1 + 1 + 2 = 4$$

etc.

Example 2:

If $f_0 = 0$, $f_1 = 1$ and $f_{n+2} = f_n + f_{n+1}$ when $n \geq 0$

We get:

f_0	f_1	f_2	f_3	f_4	f_5	f_6	f_7
0	1	1	2	3	5	8	13

This is called the Fibonacci sequence

We'll see (in chapter 8)

How to “solve” this recurrence to show

$$f_n = \left(\frac{1}{\sqrt{5}}\right)\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1}{\sqrt{5}}\right)\left(\frac{1-\sqrt{5}}{2}\right)^n$$

or

$$f_n = \left\lfloor \left(\frac{1}{\sqrt{5}}\right)\left(\frac{1+\sqrt{5}}{2}\right)^n + \frac{1}{2} \right\rfloor$$

Summations

Its often useful to compute with expressions like:

$a_3 + a_4 + a_5 + a_6 + a_7$, (add consecutive numbers from some sequence)

If this is the sequence from Reading 3, we would have:

$$\frac{1}{2} + 3 + \frac{1}{3} + \frac{3}{2} + \frac{2}{3} = 6$$

We can represent expressions like this in summation notation,

Our last sum becomes:

$$\sum_{i=3}^7 a_i$$

- 7 is the end index
- $i = 3$ is the starting index
- i is the dummy variable
- a is the name of sequence

Pseudo code:

```
S = 0 % initiate sum
```

```
for i from 3 to 7 do
```

```
    S = S + a_i
```

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next i
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```
return S
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If the start index is bigger than the end index, nothing gets added, and the result is 0.

If there is a nice formula for a_i then there might be a nice formula for the sum:

$$\sum_{i=0}^k a_i$$

Not so worried about the bottom starting index.

Example:

$$\sum_{i=0}^k i = 0 + 1 + 2 + 3 + \dots + k$$

In this case, let

$$S = 0 + 1 + 2 + 3 + \dots + k$$

so

$$S = k + (k - 1) + \dots + 0$$

thus

$$2S = k + k + \dots + k = k(k + 1)$$

hence

$$S = \frac{k(k+1)}{2}$$

Example 2:

$$T = \sum_{k=0}^n r^k = 1 + r + r^2 + \dots + r^k$$

so

$$rT = r + r^2 + \dots + r^k + r^{k+1}$$

Subtracting

$$T - rT = 1 + (r - r) + (r^2 - r^2) + \dots + (r^k - r^k) - r^{k+1}$$

so

$$(1 - r)T = 1 - r^{k+1}$$

or

$$T = \frac{1 - r^{k+1}}{1 - r}$$

Note:

If $r = 1$, then sum is just $k + 1$

More formulae

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{k=1}^n k^3 = \left(\frac{n(n+1)}{2} \right)^2$$

In chapter 5, we'll see how to verify these formulae,
in chapter 8 we'll learn how to find them

Why we don't worry about the bottom index

If we want, for example:

$$\sum_{k=9}^{15} (3k^3 - 7k^2)$$

Trick 1

$$a_9 + a_{10} + a_{11} + \dots + a_{15} = (a_1 + a_2 + \dots + a_{15}) - (a_1 + a_2 + \dots + a_8)$$

$$\sum_{k=9}^{15} (3k^3 - 7k^2) = 3 \sum_{k=9}^{15} k^3 - 7 \sum_{k=9}^{15} k^2$$

$$3 \sum_{k=1}^{15} k^3 - 3 \sum_{k=1}^8 k^3 - 7 \sum_{k=1}^{15} k^2 + 7 \sum_{k=1}^8 k^2$$

$$3 \left(\frac{15 * 16}{2} \right)^2 - 3 \left(\frac{8 * 9}{2} \right)^2 - 7 \left(\frac{15 * 16 * 31}{6} \right) + 7 \left(\frac{8 * 9 * 17}{6} \right)$$

§2.5 Cardinalities of (infinite) Sets

What can we say if two finite sets, say S and T have the same cardinality

If the size of $|S| = |T|$ then, there is some k so $|S| = k$ and $|T| = k$

This means, by rules of counting, that there is a function

$$f : S \rightarrow \{1, 2, 3, 4, 5\}$$

such that f is a bijection

f is an injection since we aren't allowed to say any number twice.

and f is a surjection since we aren't allowed to skip any numbers.

But also, there must be a bijection from T to the $\{1, 2, 3, 4, 5\}$

i.e $g : T \rightarrow \{1, 2, 3, 4, 5\}$

This means $g^{-1} \circ f$ is a bijection from S to T

For infinite sets, this is the definition of "having the same cardinality"

Definition:

Two sets S and T (possibly infinite) are said to have the same cardinality iff it is possible to find a bijection $f : S \rightarrow T$,

We write:

$$|S| = |T|$$

Similarly

$|S| \leq |T|$ iff it is possible to find an injection $g : S \rightarrow T$

This notation is justified because $=$ and $\leq$ behave as we might hope.

Theorems:

1. If $|S| = |T|$ and $|T| = |R|$, then $|S| = |R|$

This works because $f : S \rightarrow T$ is a bijection

and $g : T \rightarrow R$ is a bijection,

then $g \circ f : S \rightarrow R$ is a bijection

2. If $|S| \leq |T|$ and $|T| \leq |R|$, then $|S| \leq |R|$ (easy)

3. If $|S| \leq |T|$ and $|T| \leq |S|$, then $|S| = |T|$ (very hard) (Schroder-Bornstein)

Using this definition we find:

$$|\mathbb{N}| = |\mathbb{Z}^+|$$

but also :

$$|\mathbb{N}| = |\mathbb{Z}|$$

Surprise

$$|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}|$$